

Government of Andhra Pradesh

Women Development & Child Welfare Department

AI-Based Early Childhood Development (ECD) Innovation Challenge

1. Background and Context

Early Childhood Development (ECD) is a critical determinant of long-term health, learning outcomes and socio-economic well-being. The Integrated Child Development Services (ICDS) programme of the Women Development & Child Welfare (WD&CW) Department plays a pivotal role in delivering nutrition, health and early learning interventions to children in the age group of 0–6 years. However, existing systems rely heavily on manual observation, non-standardised assessments and delayed referrals, resulting in late identification of developmental delays, disabilities and health risks.

In view of the State's priority to adopt technology-enabled, preventive and predictive governance, and building on learnings from pilot initiatives undertaken in Andhra Pradesh, it is proposed to leverage Artificial Intelligence (AI) and digital technologies to strengthen early identification, personalised intervention and system-level decision-making within ICDS.

Accordingly, the WD&CW Department, in collaboration with **Ratan Tata Innovation Hub (RTIH)**, proposes to conduct an **AI-Based Early Childhood Development (ECD) Innovation Challenge** to identify, pilot and scale deployable AI-driven solutions.

2. Rationale for Innovation Challenge

Given the rapidly evolving nature of Artificial Intelligence, behavioral science and ECD research, a traditional RFP/EOI approach may limit innovation. A structured **Hackathon cum Innovation Challenge model**, executed as per the guidelines, enables:

- Open and competitive sourcing of diverse solutions
- Rapid prototyping and validation of ideas
- Outcome-oriented evaluation in real-world ICDS settings
- Selection of solutions with demonstrable impact for piloting and scale-up

This approach aligns with the Government of Andhra Pradesh's emphasis on innovation-led, data-driven and accountable governance.

3. Objectives of the Innovation Challenge

The Innovation Challenge aims to:

- Enable **early identification and predictive risk assessment** of developmental delays, disabilities and health risks among children aged 0–6 years;
- Support **personalised, child-specific, age appropriate and culturally contextualized intervention planning** and improved caregiver engagement;
- Strengthen **decision support, monitoring and prioritisation** for ICDS functionaries and administrators;
- Establish a **scalable, secure and government-owned digital architecture** for ECD;
- Facilitate transition from innovation to **pilot implementation and scale-up** in accordance with Government norms.

4. Scope of the Innovation Challenge

The Challenge shall invite innovative, AI-enabled solutions addressing the following structured Problem Statements:

4.1 Problem Statement I: AI-Enabled Early Identification, Risk Stratification, AI-Driven Personalized Intervention planning & Caregiver engagement

Problem Context:

Early identification of developmental delays, disabilities, defects at birth and emerging health risks such as cognitive, motor, speech-language, behavioural and socio-emotional domains among children aged 0–6 years remains inconsistent due to manual assessments, subjective observation and delayed referrals. Further, existing interventions are largely generic and not dynamically tailored to individual child risk profiles, limiting the effectiveness of corrective measures and caregiver engagement.

What is Required:

An integrated AI-enabled solution that:

- Analyses child-level growth, developmental, behavioural and health indicators to generate predictive risk scores and early alerts;
- Supports Anganwadi Workers in identifying at-risk children before visible delays become severe;
- Generates personalised, child-specific intervention plans;
- Provides context-appropriate caregiver guidance (including low-literacy formats);
- Dynamically adjusts recommendations based on child progress and updated data inputs.

Expected Outcomes

The proposed solution should clearly demonstrate the following outcomes during pilot and scale-up phases:

- 1. Objective and Standardized Screening**
 - a. Consistent developmental assessments aligned with national ECD frameworks
 - b. Reduction in subjectivity and variation across AWCs
- 2. Risk Stratification and Prioritization**
 - a. Clear classification of children into low, medium and high developmental risk categories
 - b. Automated triggers for follow-up, referral or intensified intervention
- 3. Predictive Insights**
 - a. Ability to flag children who may not yet show visible delays but are at elevated future risk
 - b. Identification of early warning patterns at individual and cohort levels
- 4. Field-Level Usability**
 - a. Minimal additional burden on Anganwadi Workers
 - b. Offline functionality and adaptability to low-connectivity environments
- 5. Individualized Intervention Pathways**
 - a. Child-specific activity recommendations based on developmental coefficient
 - b. Dynamic adjustment of interventions as the child progresses or regresses
- 6. Caregiver Empowerment**
 - a. Easy-to-understand guidance in local languages
 - b. Use of audio, visual and nudging mechanisms suitable for low-literacy households
- 7. Frontline Worker Support**
 - a. Capacity building of frontline staff
 - b. Clear guidance for AWWs on what activities to reinforce during center- based and home visits
 - c. Reduced dependency on external specialists
- 8. Demonstrable Developmental Gains**
 - a. Measurable improvement in targeted developmental domains during the pilot period

4.2 Problem Statement II: Digital Decision Support, Impact Measurement & Scalable Architecture

Problem Context:

ICDS administrators currently lack robust real-time prioritisation tools to identify high-risk clusters, performance gaps and service delivery bottlenecks. Additionally, scaling digital innovations requires strong data governance, interoperability, explain ability and measurable outcome tracking aligned with Government ownership norms.

What is Required:

A comprehensive digital ecosystem that:

- Provides role-based dashboards and analytics for Anganwadi Workers, Supervisors, CDPOs and State-level administrators;
- Enables prioritisation through alerts, clustering and performance insights;
- Incorporates measurable impact indicators for monitoring child-level and system-level outcomes;
- Ensures secure, interoperable and scalable architecture compliant with applicable data protection norms;
- Embeds audit trails and explainable AI components for transparency and accountability.

Expected Outcomes:

The solution should ensure:

1. Role-Specific Decision Support
 - a. Dashboards tailored to the responsibilities of AWWs, Supervisors, CDPOs, DW&CW&EO's and state officials
 - b. Clear visibility into child-level risks and intervention status and referral status
2. Alert and Prioritization Mechanisms
 - a. Automated alerts for high-risk children, underperforming centers or emerging trends
 - b. Suggested action pathways for each user role at AWC and Household level
3. Performance and Outcome Monitoring
 - a. Tracking of intervention delivery and workforce performance
 - b. What's app groups for household level intervention tracking
 - c. Identification of systemic bottlenecks
4. High Adoption and Ease of Use
 - a. Minimal additional data entry requirements
 - b. High usability and sustained adoption by ICDS staff
5. Longitudinal Impact Measurement

- a. Tracking of developmental outcomes over time at individual and cohort levels
 - b. Evidence generation for program effectiveness
- 6. Strong Data Governance
 - a. Consent-based data collection and role-based access controls
 - b. Compliance with Government of India data protection and cybersecurity norms
- 7. Interoperability and Government Ownership
 - a. API-based integration with ICDS and health systems
 - b. Full ownership of data and system with WD&CW
- 8. Scalability and Institutional Sustainability
 - a. Ability to scale seamlessly across all districts
 - b. Capacity building and knowledge transfer to government teams

5. Eligibility Criteria

1. **Startup Eligibility** Participation in the Innovation Challenge and eligibility for subsequent procurement shall be **restricted exclusively to startups recognised by the Department for Promotion of Industry and Internal Trade (DPIIT), Government of India**, holding a valid DPIIT Startup Recognition Certificate at the time of application.
2. **Procurement Conditions** Eligible startups shall be governed by the provisions of **G.O.Ms.No.33, Government of Andhra Pradesh**, which stipulates that:
 - A startup may be awarded a maximum of **two (2) procurement contracts per financial year**; and
 - Each procurement contract shall be subject to a maximum value of **₹1.00 crore**.

6. Procurement Structuring under G.O.Ms.No.33

the following procurement structure has been approved:

1. **Combined Procurement – Problem Statements 1 and 2** Problem Statements relating to **(i) Early Identification & Risk Stratification** and **(ii) Personalised Intervention & Caregiver Engagement** shall be combined into a single procurement proposal with a total value of **₹1.00 crore**. This shall include **capacity building and training of ICDS functionaries** as part of the Scope of Work.
2. **Combined Procurement – Problem Statements 3 and 4** Problem Statements relating to **(iii) Digital Enablement & Decision Support** and **(iv) Impact Measurement & Scalable Architecture** shall be combined into a separate procurement proposal with a total value of **₹1.00 crore**.

3. **Operations and Maintenance (O&M)** Provision for **Operations and Maintenance (O&M) for a period of five (5) years** shall be incorporated under the Innovation Challenge and subsequent procurement arrangements.

7. Hackathon Stages and Process

The Innovation Challenge shall be conducted in the following stages:

1. Issue of Hackathon Notification / EOI;
2. Submission of applications by eligible startups;
3. Screening and shortlisting by an expert committee;
4. Demonstration / pitch by shortlisted teams;
5. Final selection of pilot-ready solutions;
6. Pilot implementation in identified ICDS project areas;
7. Review of pilot outcomes and decision on scale-up or procurement.

8. Standardized Timeline and Process Flow

DESCRIPTION	TIMELINE
Launch of Hackathon and Applications starting Date	22-01-2026
Clarification/Query Session on Problem Statements	02-02-2026
Last Date for Submissions	04-03-2026
Pre-screening and validation of applications	09-03-2026
Evaluation by Subject Matter Experts & Shortlisting	13-03-2026
Demo/Pitch Day before Evaluation Committee	23-03-2026
Final Results and Announcement of Winners	27-03-2026
Post-event handholding and ground implementation support	To be Announced

9. Evaluation Criteria

Proposals shall be evaluated by an expert committee based on:

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|--|---|-----|
| • Problem–Solution Fit | – | 15% |
| • Innovation and Use of AI | – | 15% |
| • Feasibility and Scalability | – | 15% |
| • Impact on Child Development Outcomes | – | 40% |
| • Implementation Capability and Sustainability | – | 15% |

Solutions may be submitted at:

- Idea / Concept Stage
- Prototype / MVP Stage
- Pilot-ready / Deployed Stage

10. Proof of Concept (PoC) Scope

- Implement the model as Decision support layer of ICDS systems for one pilot Mandal/Block as identified by the department.
- The proof-of-concept should be tested with participation of field staff like AWWs, Supervisors, CDPOs, DW&CW&EO's to validate model accuracy and usability
- Design role based Digital Enablement platform which supports decision making of field staff
- API integration mechanism

11. Proof of Concept (PoC) Success Criteria

- Generation of actionable models and platforms with more than 95% accuracy in early childhood developmental delays.
- Multi-lingual systems that supports field level staff with complete accuracy which converts developmental insights into actionable guidance.
- Seamless integration with state dashboards for visualization and reporting.

12. Data Privacy and Compliance

The solution must ensure full compliance with the Digital Personal Data Protection (DPDP) Act, 2023, anonymize information provided by the department and adhere to sensitive children data handling protocols. Data storage and model training must occur within secure, government-approved environments.

13. Post-Demonstration Opportunity:

Following successful demonstrations before senior government officials and key stakeholders, selected applicants will be invited for further deliberations focusing on practical aspects of integration, training requirements, and regulatory compliance. Based on these discussions, a structured roadmap will be developed to facilitate pilot testing and potential large-scale adoption of the most promising innovations across the system.

14. Rewards, Pilot Support and Post-Hackathon Opportunities

- Selected startups shall receive **pilot implementation support**, subject to administrative and financial approvals;
- Winning solutions shall be eligible for **procurement under G.O.Ms.No.33**, subject to prescribed limits and conditions;
- Technical mentoring, field facilitation and Government handholding shall be provided during the pilot phase;
- Based on demonstrated outcomes, solutions may be considered for **scale-up and institutionalisation** within ICDS.

15. Conclusion

The proposed AI-Based ECD Innovation Challenge is a strategic initiative to strengthen early childhood outcomes through responsible use of AI, while ensuring government ownership, data security and long-term sustainability. Transitioning from a conventional EOI/RFP to an **Innovation Challenge model** positions Andhra Pradesh at the **forefront of technology-driven child development governance**.